

BALÁZS VARGA, PHD

Modelling & Simulation | Optimization | Applied ML | Mobility & Physical Systems
Budapest, Hungary | b.varga.90@gmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Scholar](#)

PROFILE

Applied scientist with a strong engineering background in modelling, optimization, control and applied ML, experienced in turning hard engineering problems into implemented and validated solutions.

SELECTED TECHNICAL PROJECTS

Nationwide Real-Time Traffic Estimation 2026–present

Applied Scientist / Technical Owner | KJ group, Georgia; commercial mobility platform

- Owned the R&D and implementation of an end-to-end, real-time traffic estimation system for a nationwide road network, from raw vehicle trajectories and map matching through ML-based inference and multi-source data fusion.
- Designed the data and graph-processing pipeline using Python, Polars, Valhalla, OpenStreetMap and XGBoost, including historical fallback logic, confidence handling and road-closure-aware estimation.
- Achieved SQV > 85% and delivered the system for integration into the client's production platform.

TrafficSim – Automated Intersection Analysis SaaS 2025–present

Co-founder / Technical Lead | Self-funded engineering SaaS

- Co-founded and lead the technical development of an engineering SaaS that automates intersection modelling, traffic simulation and standards-based capacity analysis.
- Built the core simulation and calculation backend as a Python microservice, integrating and adapting the open-source SUMO simulator for automated intersection analysis.
- The product is actively evolving based on practitioner feedback, with additional regulatory methods and international distribution under development.

ZalaZONE – Mixed-Reality Digital Twin for Autonomous Vehicle Testing 2018–2019

Research Engineer / Lead Developer for Orchestration | Autonomous-vehicle validation and cyber-physical testing

- Developed the central orchestration and co-simulation layer of a mixed-reality vehicle-testing platform, synchronizing real vehicles with simulated traffic, virtual objects and CAN-based sensor information.
- Integrated traffic simulation, virtual-object handling and physical vehicle interfaces into automated-driving test scenarios.
- The system was validated in multiple public demonstrations for automated-driving use cases.

Scalable Mixed-Reality Validation Research Programme 2022–present

Principal Researcher | BME / National Laboratory for Autonomous Systems

- Develop methods for computationally scalable mixed-reality validation of connected and automated cyber-physical systems, combining adaptive multi-fidelity co-simulation, simulation optimization and design of experiments (DoE).
- Developed and validated a traffic–vehicle co-simulation method that retained high fidelity around the system under test while dynamically downscaling distant traffic; achieved a 200–500% simulation-performance improvement while retaining modelling accuracy.
- Technically supervised V2X mixed-reality extensions, including VIL experiments; currently extending the programme toward traffic–CFD coupling and physics-informed neural networks (PINNs) for multi-physics validation.

EMPLOYMENT

Budapest University of Technology and Economics (BME) 2017–present

Researcher. Applied R&D in modelling, simulation, optimization, control; selected industry collaborations: Bosch, Google Research, aiMotive, ZalaZONE, Főnterv; Developed and teach MSc/PhD courses in computational modelling and numerical methods; supervise MSc and doctoral research.

AVL Hungary 2016–2018

Function Developer. Model-based development and SIL validation of heavy-duty Euro 6 diesel engine and aftertreatment functions in MATLAB/Simulink.

Chalmers University of Technology 2015; later postdoctoral appointment

Project Assistant / Postdoctoral Researcher. Vehicle systems, traffic simulation, control and reinforcement-learning research.

HUN-REN SZTAKI 2012–2014

Research Intern. Vehicle-dynamics modelling and control, including active anti-roll and variable-geometry suspension systems.

TECHNICAL EXPERTISE

Applied ML: Gaussian Processes · Neural Networks · Reinforcement Learning · XGBoost

Optimization: Constrained Optimization · Multi-objective Optimization · Genetic Algorithms · Design of Experiments

Control: Model Predictive Control · Robust Control · Linear Parameter Varying Control

Computational Modelling: Physics-based Modelling · Multi-fidelity Methods · Co-simulation · Real-time Simulation

Programming & Software Engineering: Python · PyTorch · MATLAB/Simulink · Rust · PHP · Docker · Git · Microservices

EDUCATION

PhD, Transportation / Control Engineering – BME 2017–2020

Model predictive control for public-transport operations and network-level control.

MSc, Automotive Engineering – BME / Chalmers University of Technology 2013–2015

MSc thesis with Volvo Trucks on LPV steering control for long heavy vehicle combinations.

BSc, Vehicle Engineering – BME 2009–2013

Vehicle dynamics and control.

LANGUAGES

Hungarian: Native | **English:** Fluent / full professional proficiency | **German:** Intermediate

SELECTED PUBLICATIONS

- **B. Varga et al.**, “Deep Q-learning: A robust control approach,” *International Journal of Robust and Nonlinear Control*, 2023.
- **B. Varga et al.**, “Optimizing vehicle dynamics co-simulation performance by introducing mesoscopic traffic simulation,” *Simulation Modelling Practice and Theory*, 2023.
- **B. Varga et al.**, “Data-driven distance metrics for Kriging – Short-term urban traffic state prediction,” *IEEE Transactions on Intelligent Transportation Systems*, 2023.
- **B. Varga et al.**, “On the relationship of speed limit and CO₂ emissions in urban traffic,” *Transportation Research Interdisciplinary Perspectives*, 2025.
- **B. Varga et al.**, “Optimally combined headway and timetable reliable public transport system,” *Transportation Research Part C: Emerging Technologies*, 2018.